

Tutorial 1

1. Consider the preference relation $\geq$ defined by $(x_1, x_2) \geq (x'_1, x'_2)$ if and only if either (i) $x_1 > x'_1$ or (ii) $x_1 = x'_1$ and $x_2 \geq x'_2$. Is this preference relation complete, transitive, and increasing?

Answer: $\geq$ is complete, transitive, and increasing.

Complete: Let $x = (x_1, x_2)$ and $x' = (x'_1, x'_2)$ be two bundles. If $x_1 > x'_1$ then $x \geq x'$ if $x'_1 > x_1$ then $x' \geq x$. If $x_1 = x'_1$ and $x_2 \geq x'_2$ then $x \geq x'$. If $x_1 = x'_1$ and $x'_2 \geq x_2$ then $x' \geq x$. As these situations are exhaustive it must be that either $x \geq x'$ or $x' \geq x$ as claimed.

Transitive: Suppose $x \geq x'$ and $x' \geq x''$. This means either (a) $x_1 > x'_1$ or (b) $x_1 = x'_1$ and $x_2 \geq x'_2$ and also either (i) $x'_1 > x''_1$ or (ii) $x'_1 = x''_1$ and $x'_2 \geq x''_2$. In case (ai), (aii), and (bi) we have $x_1 > x''_1$ and so $x \geq x''$. In case (bii) we have $x_1 = x''_1$ and $x_2 \geq x''_2$ and so $x \geq x''$.

Increasing: Suppose $x \geq x'$. This means either $x_1 \geq x'_1$ and $x_2 \geq x'_2$ and thus $x \geq x'$.

2. Draw the budget line for prices $(1, 2)$ and $(2, 1)$ when expenditure is 2. Draw the consumption bundle $x = (1, 1)$. Is this bundle affordable at either budget?

Answer: Just draw the budget as I did in the lecture slides. Bundle x is not affordable. In particular, $(1, 2) \cdot (1, 1) = (2, 1) \cdot (1, 1) = 3 > 2$

3. Suppose a preference maximizing consumer in period 1 buys $x^1 = (2, 2)$ when prices are $p^1 = (1, 1)$ and $x^2 = (0, 3)$ when prices are $p^2 = (2, 1)$. In which period is the consumer better off (i.e. in which period does the consumer obtain a better bundle)?

Answer: We have $x^1 P x^2$ and $x^2 \not\prec x^1$ and so the consumer prefers x^1 to x^2 . The consumer was better off in period 1.

4. Consider a consumer who buys $x^1 = (1, 2)$ when prices are $p^1 = (1, 2)$ and $x^2 = (2, 1)$ when prices are $(2, 1)$. The consumer does not satisfy WARP. Prove that there is no increasing preference $\geq$ which can explain the consumer's behavior. That is, show that

there is no increasing $\geq$ so that $x^1 \geq x$ for all $x \in B(p^1, p^1 \cdot x^1)$ and $x^2 \geq x$ for all $x \in B(p^2, p^2 \cdot x^2)$.

Answer: Let $\geq$ be an increasing preference which satisfies $x^1 \geq x$ for all $x \in B(p^1, p^1 \cdot x^1)$. Let $\bar{x} = (3, 1)$. As $p^1 \cdot x^1 = 5 = p^1 \cdot \bar{x}$ we have $x^1 \geq \bar{x} > x^2$ and so $x^1 > x^2$ (in particular, $x^2 \not\geq x^1$). As $p^2 \cdot x^2 = 5 > 4 = p^2 \cdot x^1$ we see that $x^1 \in B(p^2, p^2 \cdot x^2)$ but $x^2 \not\geq x^1$.

5. Define a non-increasing preference $\geq$ which can explain the behavior of the consumer in the previous question.

Answer: here's a boring answer. Let $\geq$ be defined by $x \geq x'$ for all $x, x' \in \mathbb{R}_+^L$.

6. The first maxim of classic consumer theory states that "What is chosen is preferred to what could have been chosen." Can you think of some situations where this maxim may be problematic?

Answer: there are many examples where people might not choose what they prefer. For instance, they have limited attention. There are too many choices for them to consider all options. Think choosing a breakfast cereal from 20 options. People may also not know their true preferences for products they have never tried before. People may have self-control issues. For instance, they want to get in-shape but keep buying ice cream.